

Deep Reinforcement Learning

Q-learning $\rightarrow$ Q-table $Q(s,a) \Rightarrow$ 48 states
(discrete, small)

Deep Q-Networks (DQN)

problem

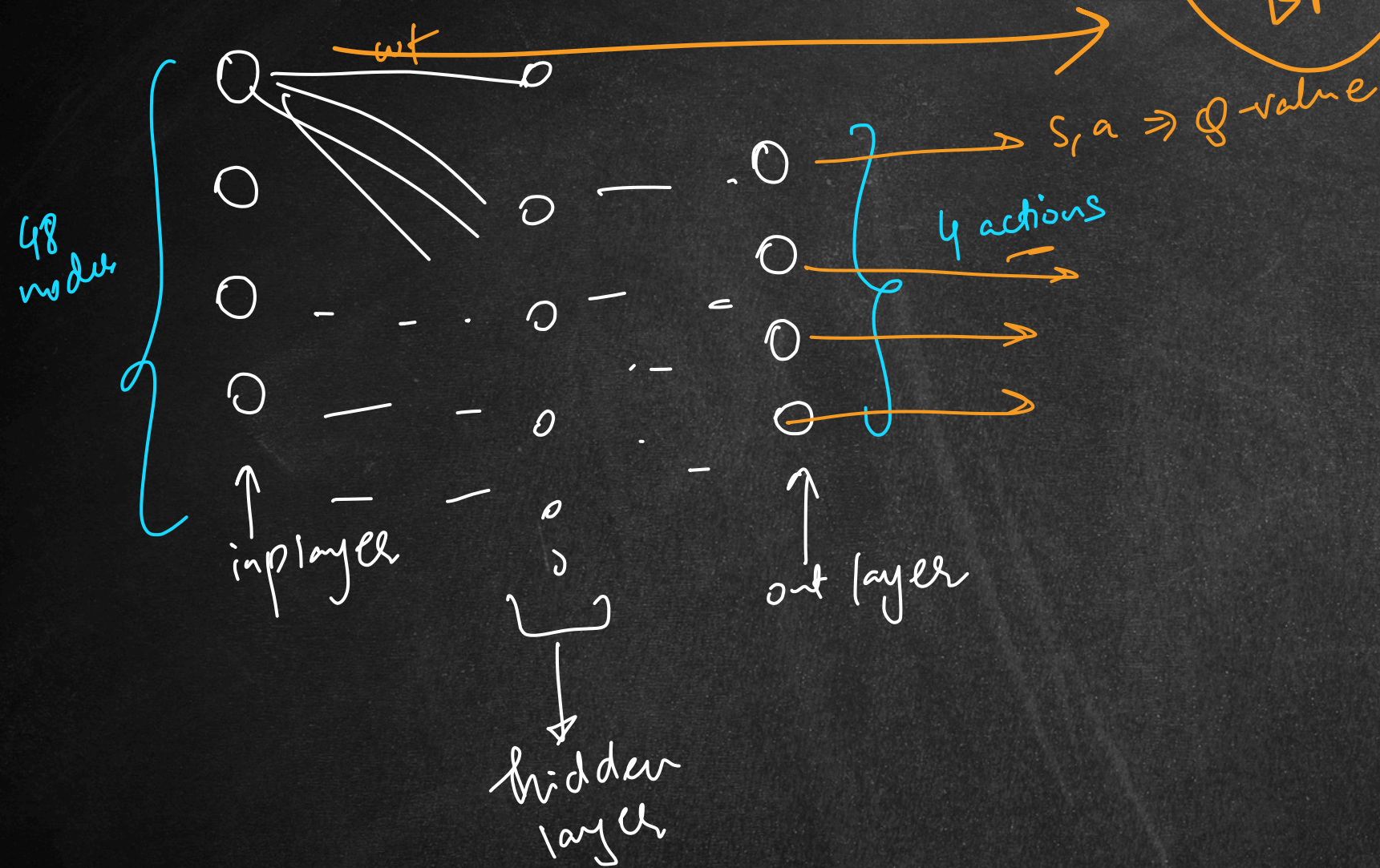
- ① millions of states $\rightarrow$ huge
- ② continuous state val $\rightarrow$ impossible

state $\rightarrow$ ~~NN~~ function $\rightarrow$ actions
Q-val

approximate or predict

Deep Q-Networks

DQN



48 states × 4 action → Q-value Q-fast

APNA
COLLEGE

Q-learning ⇒ $Q(s, a)$

DQN ⇒ $Q(s, a; \theta)$
↓ NN parameters

① Correlated samples

② Moving Target

Deep Q-Networks

Naive DQN

Correlated Samples & Experience Replay

experiences = State, Action, Reward, Next State, Terminated

batch size = 16

Problem
NN $\rightarrow$ independent

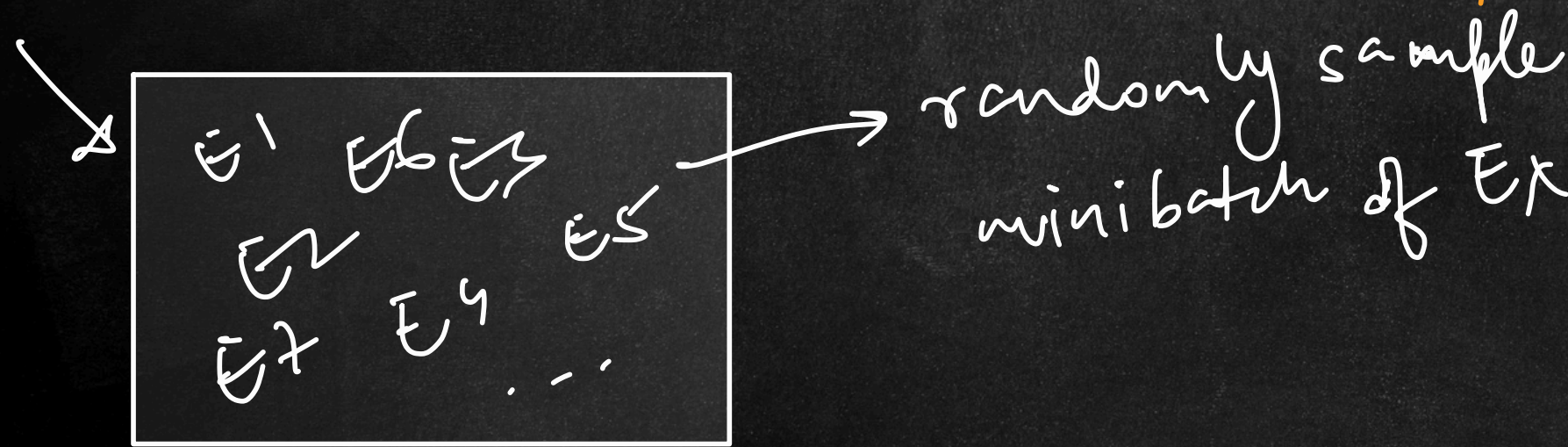
- ① sequential
- ② correlated

E1: $S1 \rightarrow A1 \rightarrow R2 \rightarrow S2$
E2: $S2 \rightarrow A2 \rightarrow R3 \rightarrow S3$
E3: $S3 \rightarrow A3 \rightarrow R4 \rightarrow S4$

X $E1 \rightarrow E2 \rightarrow E3 \rightarrow E4 \dots$
✓ $E9 \rightarrow E121 \rightarrow E2 \rightarrow E5$

$E1 \rightarrow E2 \rightarrow E3 \dots$

Solⁿ: ER $\Rightarrow$ DQN memory (replay buffer)



Deep Q-Networks

Moving Target & Target Network

leads to unstable training

Q-learning update $\Rightarrow Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_a Q(s',a) - Q(s,a)]$

target $\Rightarrow y = r + \gamma \max_a Q(s',a;\theta) \rightarrow y$

predict Q-value = $Q(s',a;\theta) \rightarrow \hat{y}$

loss = $(y - \hat{y})$

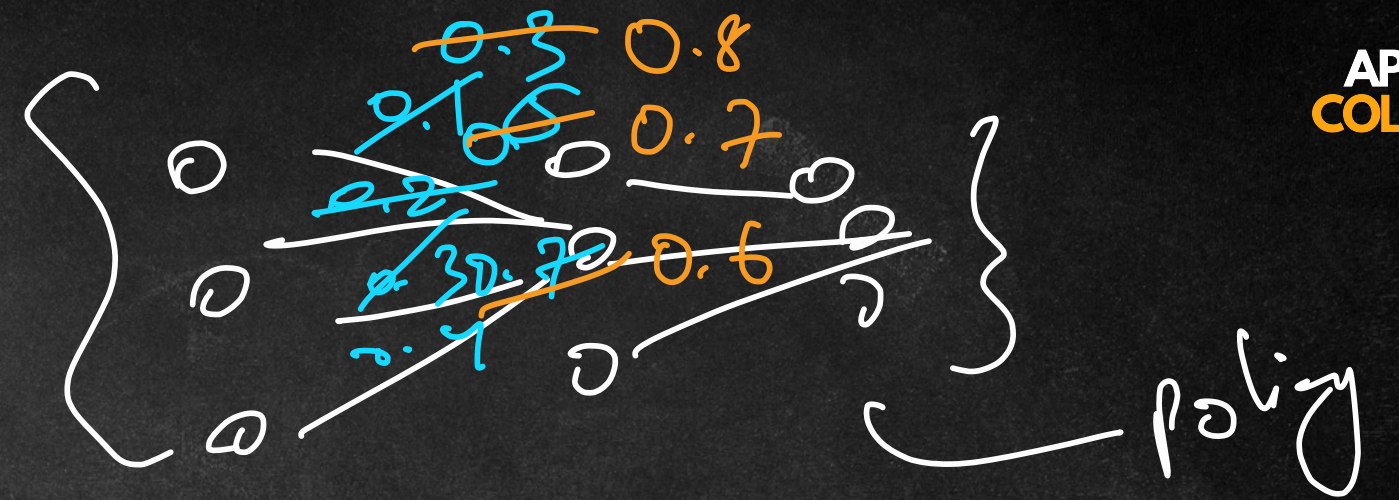
target is not a fixed value



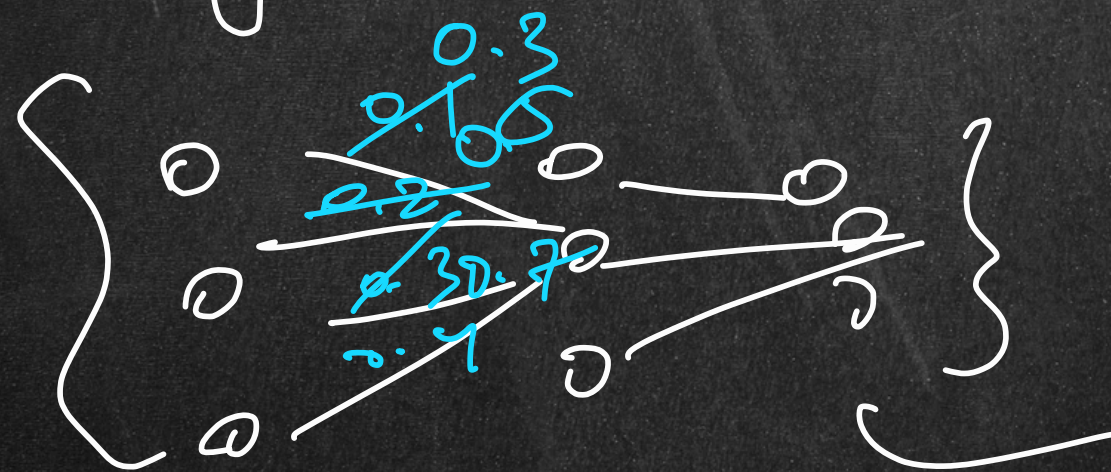
Deep Q-Networks

Moving Target & Target Network

Naive DQN $\rightarrow$ 1NN



- DQN
- ① Policy Network (online)
 - ② Target Network (old copy of policy network)



Architecture
Single

Deep Q-Networks

Training Flappy Bird

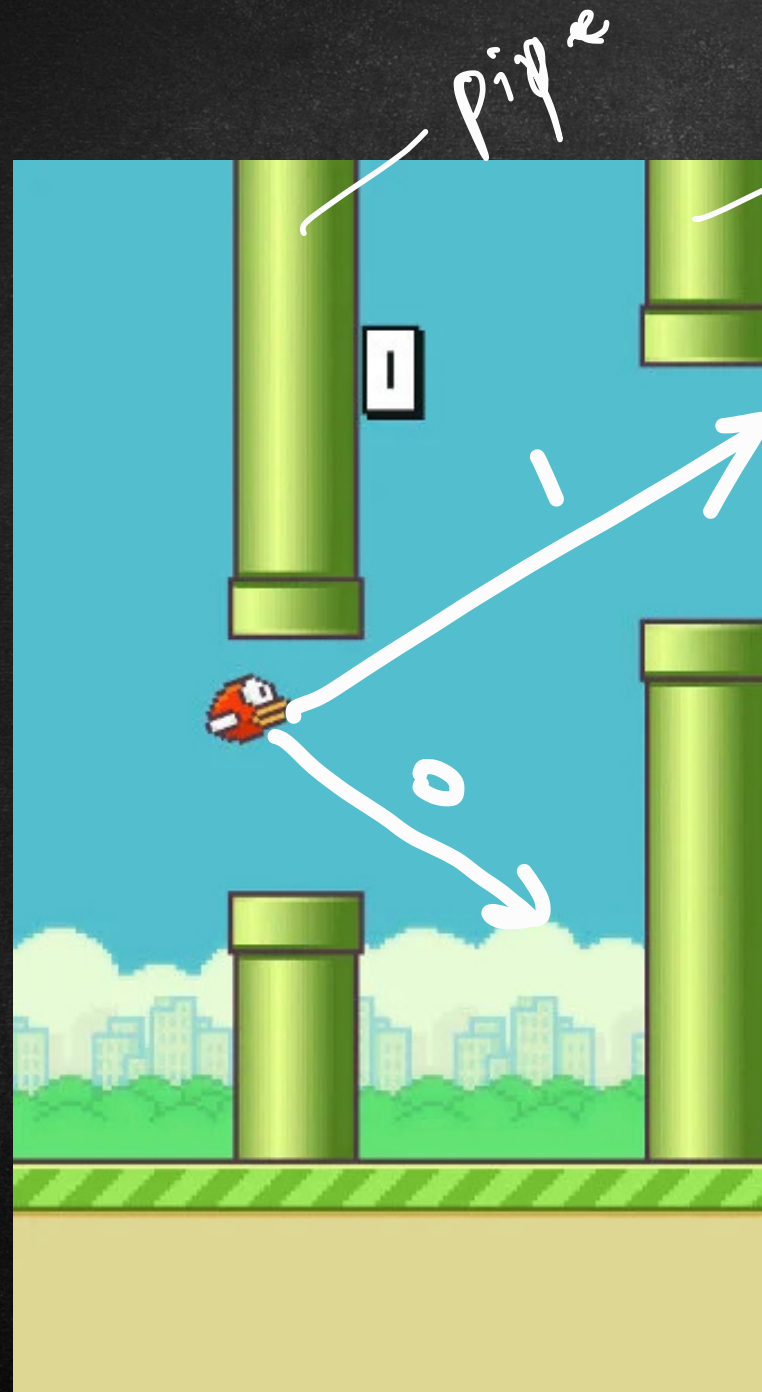
Deep Q-Networks

Training Flappy Bird

Agent

Reinforcement learning
mines Project

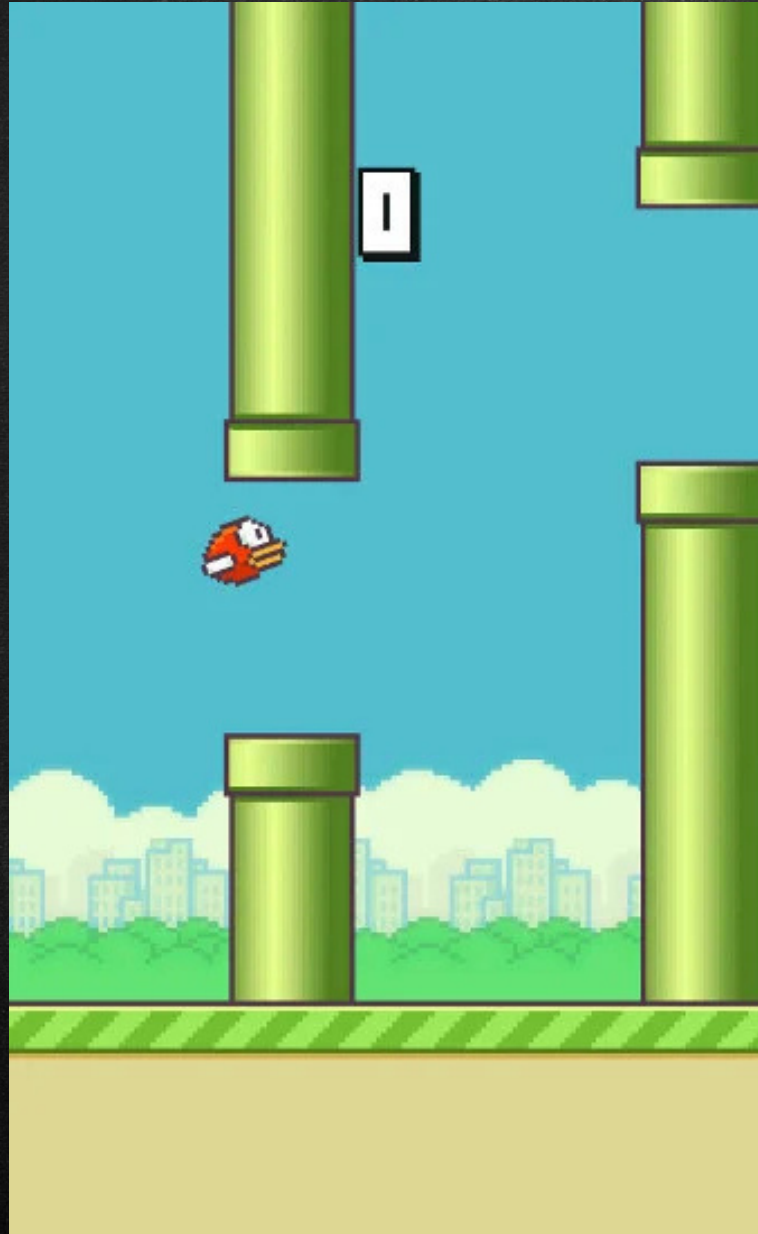
APNA
COLLEGE



0 → do nothing
1 → flap

Deep Q-Networks

Training Flappy Bird



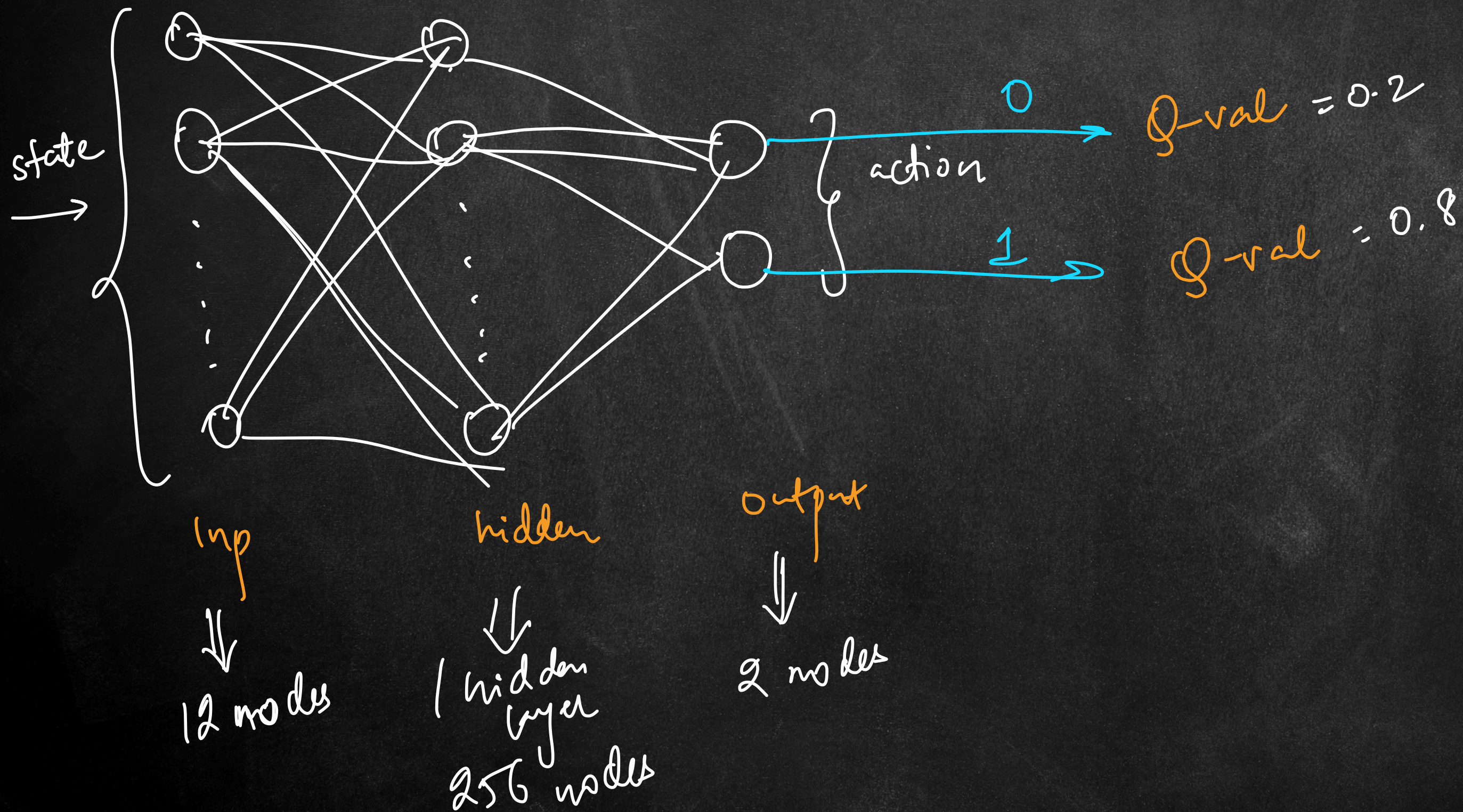
LIDAR

$E = 1$
↓
0.9
↓
0.8
↓
0.7
⋮
↓
0.05

- ① Set Env
- ① Define DQN
- ② Create the Experience Replay
- ③ multiple episodes DQN train → setup
- ④ Epsilon-greedy policy + hyperparameters
(epsilon decay)
- ⑦ Target Network
- ⑧ Train & Test

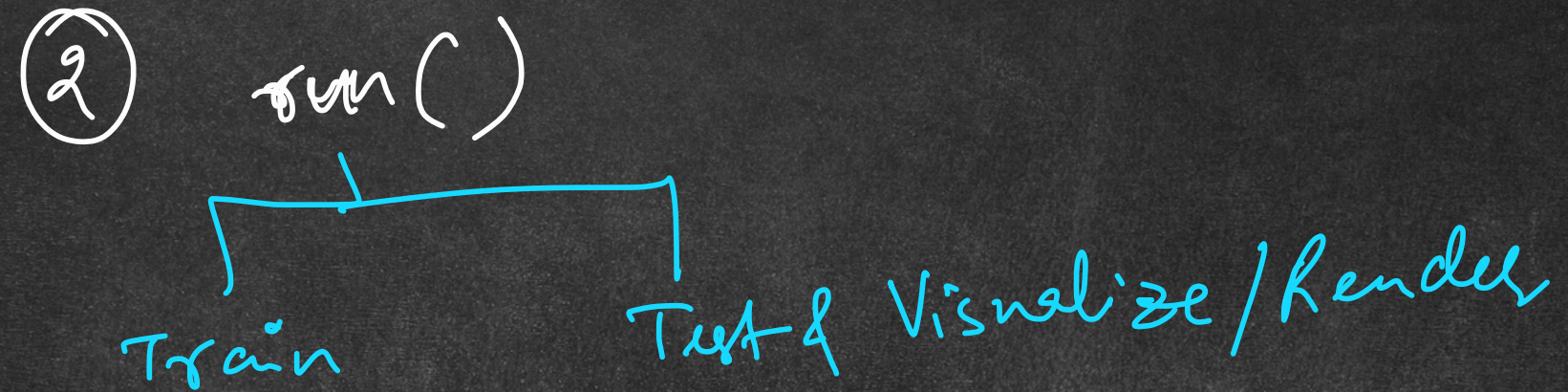
Deep Q-Networks

Training Flappy Bird $\rightarrow$ DQN $\rightarrow$ neural network arch



Deep Q-Networks

Training Flappy Bird



Deep Q-Networks

Training Flappy Bird

↓
Experience Replay

Deque (double ended queue)

add/remove

add/remove

add: 1, 2, 3, 4

remove

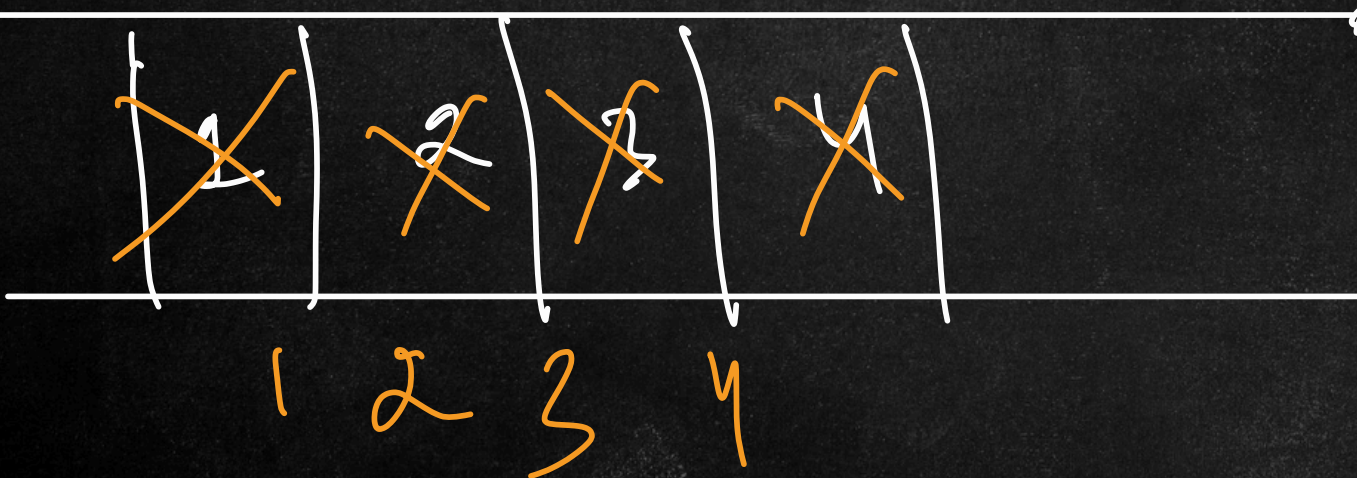
FIFO mode — First In First Out

LIFO — Last In First Out

→ stack

rear end
add

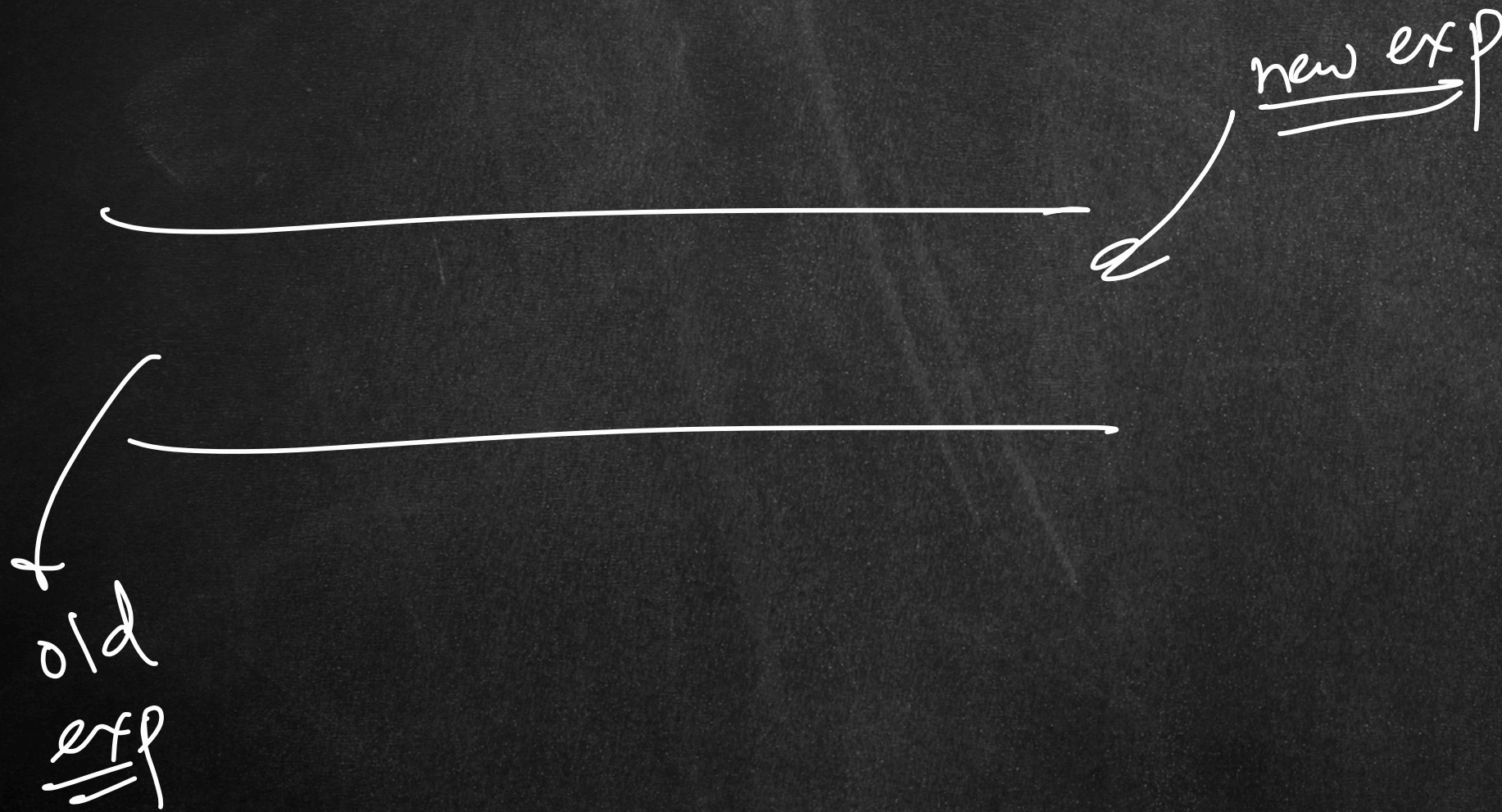
remove



Deep Q-Networks

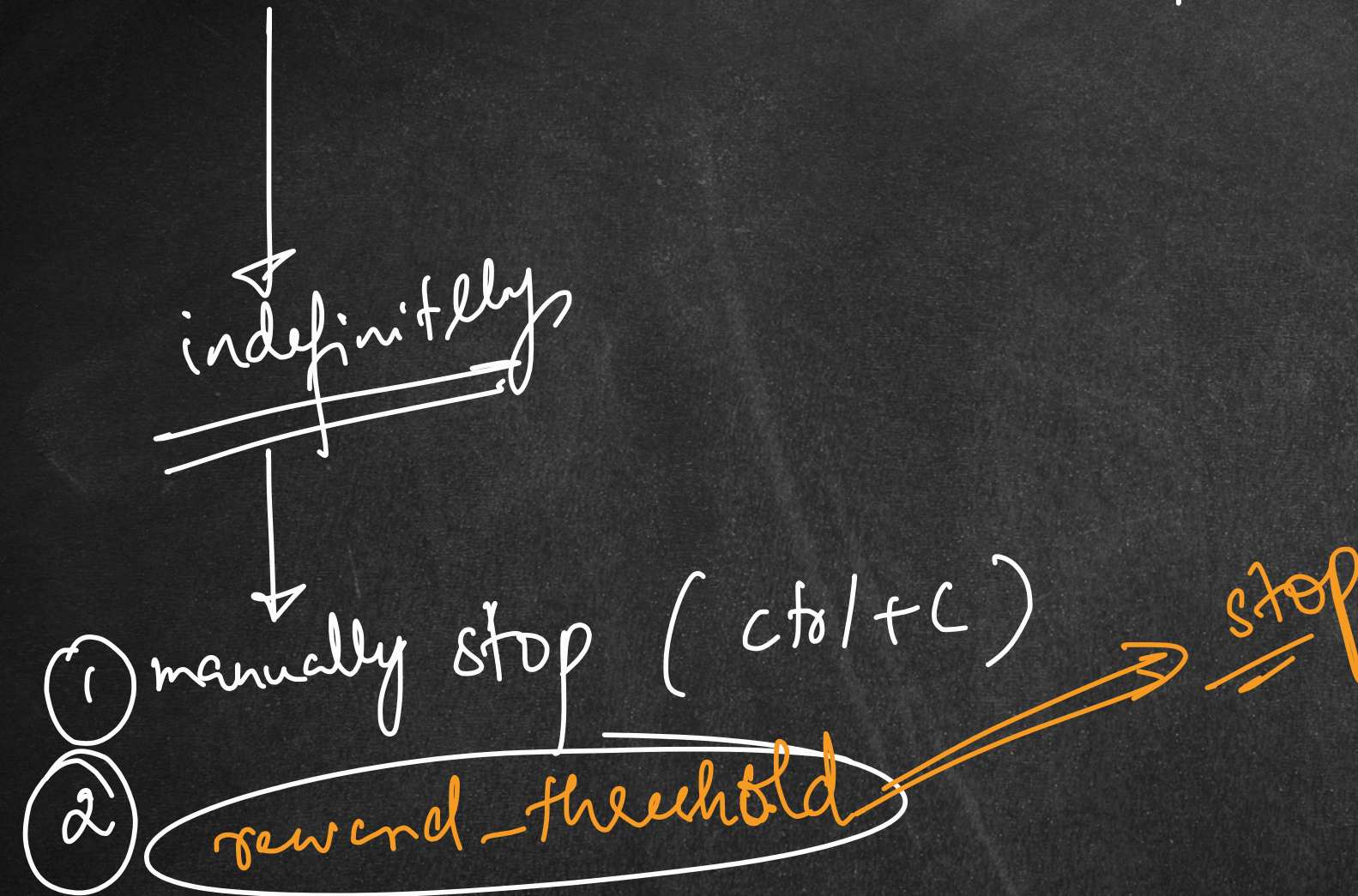
Training Flappy Bird

experience : State, Action, Reward, Next State, Terminated

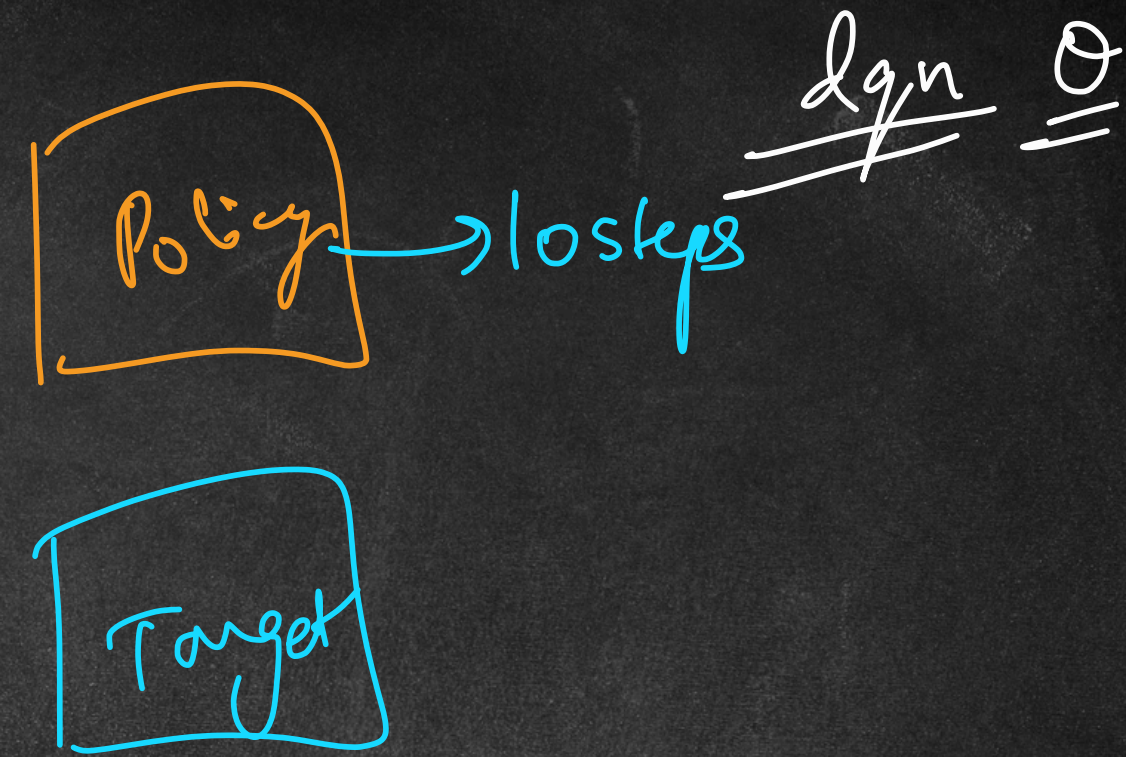


Deep Q-Networks

Training Flappy Bird



2000 ep
5000 ep



Deep Q-Networks

Training Flappy Bird

→ ϵ - greedy policy

→ ϵ - decay

→ tensors